

PrecipQC

Gauge Reading Image Quality Checker

User Manual

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This manual explains how to prepare the input data and use the interface to verify cup rain gauge readings against photographic records.

Browser required: Google Chrome

1. OVERVIEW	3
2. PREPARING THE INPUT FILES	3
2.1 Totaling gauge CSV (required)	3
2.2 Bucket CSV (optional)	4
2.3 Image Folder	5
3. OPENING THE TOOL	5
4. LOADING YOUR FILES (SETUP SCREEN)	6
4.1 Load the Totaling Gauge CSV	6
4.2 Load the Bucket CSV (optional)	6
4.3 Select the Image Folder	6
4.4 Select dates to review	7
4.5 Select values to review	7
4.6 Starting the Review	7
5. THE REVIEW SCREEN	8

5.1 The Top Bar	8
5.2 The Left Panel — Record Information	9
5.3 Making a Decision	9
5.4 The Notes Field	10
5.5 Navigating Between Records	10
5.6 The Image Panel.....	11
6.1 Output Files.....	12
7. TROUBLESHOOTING	12
ANNEX I: PREPARING INPUT FILE FOR TOTALING GAUGE READINGS.....	13
ANNEX II: PREPARING INPUT FILE FOR TIPPING BUCKET READINGS	18
Method A: Python Script	19
Method B: Microsoft Excel Only	23

1. Overview

The PrecipQC tool is a web-based application that helps you check whether the totaling rain gauge readings recorded in your data match the photographs taken at each station. It runs entirely in your web browser - no internet connection is required.

The tool works by showing you, for each station and date, three things side by side:

- The value recorded in your CSV file (totaling gauge reading)
- The tipping bucket reading for the same date, if available
- The photograph(s) of the gauge taken on that date

This guide guides you through the workflow, which mainly involves:

1. Preparing the input file for totaling gauge readings
2. Preparing the input file for tipping bucket readings
3. Manually checking each reading using the web app

You should have the following files inside the precip-qc.zip folder. Unzip the folder, and save the raw files from the ODK for totaling gauges and the tipping bucket in the same folder.

Name

	bucket_aligned.py
	precip_qcv5_manual_1.1.pdf
	PrecipQC_v5.html
	README.md
	Totalling_Gauge_Processing.xlsm

2. Preparing the Input Files

You need to prepare two CSV files before opening the tool.

2.1 Totaling gauge CSV (required)

File preparation is most easily done by using a macro, which runs within Excel and automatically applies the following instructions. See [Annex I](#). This file contains your totaling rain gauge readings and the filenames of the associated photographs. Every row represents one reading at one station on one date.

File preparation can also be done by manually renaming columns, formatting numbers and removing redundant columns in Excel. The file must have exactly these five columns, with these exact names in the first row:

Column name	Example	Notes
date	2026-02-20	The date of the reading. Change the format to “YYYY-MM-DD” if necessary.

location	Garizegu	Name of the station. Spelling must be consistent across all rows. Capital letters do not matter — the tool will match them automatically. Rename 'uds_wacwisa' to 'Kyankpala'.
gauge_mm	14.5	The totaling gauge reading in mm. Use a decimal point, not a comma (e.g. 14.5 not 14,5). For zero rain, enter 0.
image1	1771572962720.jpg	The filename of the first image, including .jpg at the end. Do not include any folder path, just the filename.
image2	1771571902373.jpg	The filename of a second image, if a second photo was taken for the same reading. Leave this cell blank if there is only one image. More image columns are possible.

	A	B	C	D	E
1	date	location	gauge_mm	image1	image2
2	2026-02-20	Garizegu	0	1771572962720.jpg	
3	2026-02-20	Gbullung	0	1771571879	17715719023
4	2026-02-20	Sanga	0	1771570846639.jpg	
5	2026-02-20	Tuunaayili	0	1771569753055.jpg	
6	2026-02-19	Tingoli	0	1771495798341.jpg	
7	2026-02-19	Sanga	0	1771485626588.jpg	
8	2026-02-19	Garizegu	0	1771485579101.jpg	
9	2026-02-19	Gbullung	0	1771485223	17714852426
10	2026-02-19	Tuunaayili	0	1771482602037.jpg	
11	2026-02-18	Tingoli	0	1771414356891.jpg	

Figure 1. Example of a totaling gauge input CSV file

2.2 Bucket CSV (optional)

File preparation is most easily done using the Python script (please refer to [Annex II](#)). This file contains the tipping bucket logger data. It is optional. If you do load it, the tool will display the corresponding tipping bucket reading alongside the totaling gauge value for each record, which can help you identify suspicious entries.

The file must have exactly these three columns:

Column name	Example	Notes
location	GARIZEGU	Name of the station in capital letters. The tool will match this to the gauge data regardless of capitalisation, so GARIZEGU will match Garizegu.
date	2026-02-20	The date of the reading. Must follow the same format as the gauge CSV.
bucket_mm	7.2	The total rainfall in millimetres calculated from the tips. Number of tips × 0.2 mm per tip.

Important Note About Date Cutoff for Tipping Bucket Data

The tipping bucket logger records continuous UTC timestamps for all rainfall events. To ensure full alignment with the manual totaling gauges—which are read and emptied each morning (typically at 07:00 UTC), with missing days defaulting to 07:00 UTC—the processing workflow **uses dynamic, observation-aligned time intervals**.

This means:

- Daily rainfall totals for the tipping bucket are calculated using strict half-open intervals: [interval start, interval end)
- Each interval is defined by the **actual daily reading time of the totaling gauge** for that location and day
- When a gauge reading is missing, the interval uses a standard 07:00 UTC cutoff for consistency
- The interval spans from the prior day's reading time (or 24 hours prior for the first record) to the current day's reading time
- All tipping bucket events falling within this custom interval are assigned to the **current observation date**

This method ensures the tipping bucket daily totals perfectly match the real-world collection schedule of the totaling gauges, creating identical date and time definitions across both rainfall datasets for accurate comparison.

Note: Days where there is only tipping bucket data and no totaling gauge reading will be automatically excluded from the review. Only records with at least one photograph are included.

2.3 Image Folder

All your photographs must be stored in a single folder on your computer. The filenames in the images must match exactly what is written in the image1 and image2 columns of the gauge CSV, including the .jpg extension.

3. Opening the Tool

The tool is a single file named PrecipQC.html. You do not need to install any software to use it.

1. Locate the file PrecipQC.html on your computer.
2. Right-click the file and choose **Open with** → **Google Chrome**. Do not open it with Firefox, Safari, or Internet Explorer.
3. The setup screen will appear in your browser window.

PrecipQC_v5
Gauge reading image quality checker - Load your files to begin

01 **Totaling gauge CSV** No file selected

02 **Bucket CSV (optional)** No file selected

03 **Image Folder** No folder selected

04 **Select dates to review**
START DATE: 03/01/2025 END DATE: 06/12/2026

05 **Totaling Gauge Filter (optional)**
All data (include zeros)

Figure 2 The setup screen

4. Loading Your Files (Setup Screen)

When the tool opens, you will see a setup screen with three steps.

4.1 Load the Totaling Gauge CSV

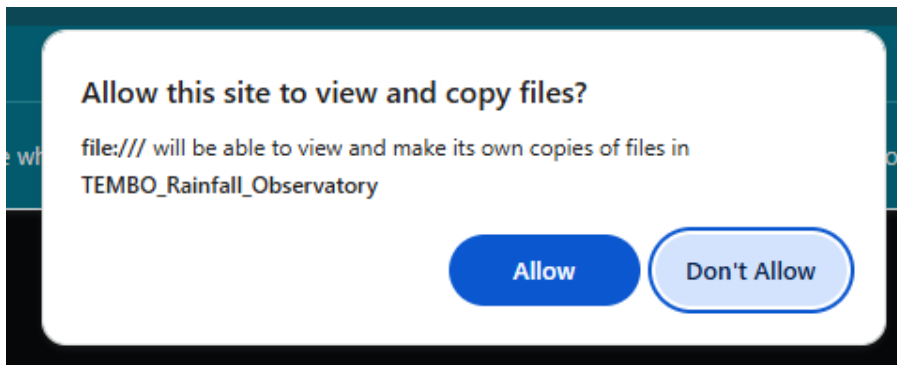
1. Click the **Browse** button next to **Totaling gauge CSV**.
2. A file picker window will open. Navigate to your gauge CSV file (eg. totaling_input_XXXX.csv) and click Open.
3. The filename will appear below the button, and a green tick (✓) will appear on the right to confirm it has been loaded successfully.

4.2 Load the Bucket CSV (optional)

4. Click **Browse** next to **Bucket CSV** if you have tipping bucket data.
5. Select your bucket CSV file (eg. bucket_aligned_XXXX.csv) and click Open.
6. If you do not have bucket data, you can skip this step entirely. The tool will still work — the “Tipping bucket” field will show **No data** for all records.

4.3 Select the Image Folder

7. Click **Browse** next to **Image Folder**.
8. A folder picker window will open. Navigate to the folder containing your photographs and click Select Folder.
9. Your browser may ask for permission to access the folder — click Allow or Yes.



10. The folder name will appear below the button with a green tick.

4.4 Select dates to review

11. If desired, select start and end dates for what you want to review in this session.

4.5 Select values to review

12. The default option is to review all records from the totaling gauge, where there is a photo and/or a tipping bucket value to verify. You can also choose to work on only zero values or only non-zero values from the dropdown list.

4.6 Starting the Review

Once the Totaling gauge CSV and Image Folder are both loaded (the Bucket CSV is optional), the **Start Review** → button at the bottom will become active. Click it to begin.

PrecipQC_v5
Gauge reading image quality checker · Load your files to begin

01 **Totaling gauge CSV *** ✓
totaling_input_20260612_151948.csv

02 **Bucket CSV (optional)** ✓
bucket_aligned_2026-06-12_15-01-09.csv

03 **Image Folder *** ✓
media

04 **Select dates to review ***
START DATE: 02/21/2026 END DATE: 06/12/2026

05 **Totaling Gauge Filter (optional)** ○
All data (include zeros)

Figure 3 Sample of a setup screen ready for review

5. The Review Screen

After clicking Start Review, the main review screen opens. It is divided into two parts: a panel on the left showing the data for the current record, and a large image area on the right showing the photograph(s).

5.1 The Top Bar

The top bar runs across the full width of the screen and shows three things at all times:

- **Progress bar and counter** — shows how many records you have reviewed out of the total (e.g. 47 / 350). The blue bar fills up as you confirm decisions.
- **Filter button** — opens a panel to select specific stations to review. Only records from your chosen stations will appear in the workflow, while preserving all existing decisions.
- **Jump to button** — lets you quickly navigate to any record by entering a full or partial date (YYYY-MM-DD), skipping directly to the first matching entry.
- **Colour tallies** — three small badges showing the running count of MATCH (green), UNSURE (yellow), and WRONG (red) decisions made so far.
- **End & Export button** — in the top-right corner. Click this at any time to finish the session and download your output files.

5.2 The Left Panel — Record Information

The left panel shows the data for the record currently on screen.

Record section

- **Station** — the name of the monitoring station.
- **Date** — the date of the reading.

Readings section

- **Totaling gauge (mm)** — the value recorded in the gauge CSV for this station and date. This is the number you are checking against the photograph.
- **SPLIT READING badge** — if this appears, the tool has detected multiple rows for the same station and date. For example, when the gauge overflowed and was emptied before a second reading was taken for the remaining water. The individual sub-readings (e.g. Row 1: 26.0 mm, Row 2: 14.0 mm) are shown below the badge. The highest value is displayed by default.

READINGS		
Cup gauge (mm)	12 mm (split)	
	SPLIT READING	
Row 1	0.0 mm	1759496237791.jpg
Row 2	12.0 mm	1759394922322.jpg
Tipping bucket (mm)	No data	
Discrepancy	—	

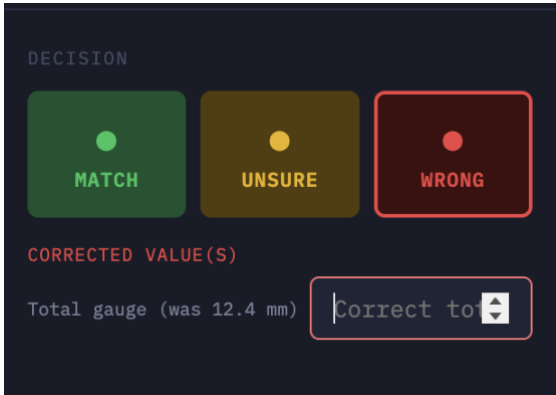
- **Tipping bucket (mm)** — the corresponding tipping bucket reading, if available. Shows “No data” if no bucket data was loaded or if there is no record for this date.
- **Discrepancy** — the difference between the totaling gauge and bucket readings (totaling gauge minus bucket). Shown in green if small, yellow if moderate, and red if large. A large discrepancy does not necessarily mean the reading is wrong, but it is a useful flag to look more carefully at the image.

Already-reviewed banner

If you navigate back to a record you have already confirmed, a coloured banner will appear showing the decision you recorded (e.g. “✓ Confirmed: Match”). You can change the decision by clicking a different button and confirming again.

5.3 Making a Decision

After examining the photograph, press one of the three decision buttons:

Button	Meaning	When to use and action needed
MATCH (number key 1)	The image clearly shows the totaling gauge, and the value in the CSV matches what you see in the photo.	/
UNSURE (number key 2)	You cannot reach a confident conclusion. The record will be excluded from the clean dataset.	Image is blurry, too dark, gauge is cut off, or there is a large and unexplained difference between the tipping bucket and totaling gauge readings.
WRONG (number key 3)	The image is clear and readable, but the value recorded in the CSV is different from what the photo shows.	You can clearly read the gauge in the photo, but the number does not match the CSV. Enter the correct value in the box that appears. 

Confirming your decision

Once you have pressed a decision button (and entered a corrected value if applicable), click the **Confirm** → button at the bottom of the left panel. You can also press **Enter** on your keyboard instead of clicking. After confirming, the tool automatically moves to the next record.

Note: Your decision is not saved until you click Confirm (or press Enter). Pressing a colour button alone does not save anything.

5.4 The Notes Field

Below the decision buttons is an optional Notes field. You can type any observations here — for example, why you marked something as UNSURE, or a specific detail you noticed in the photograph. Notes are saved in the training_notes.json output file and are useful for future reference, or for training for AI tools. The Notes field is completely optional. You can leave it blank for any or all records.

5.5 Navigating Between Records

Use the ← **Prev** and **Next** → buttons at the bottom of the left panel to move between records. You can go back to review or change a previous decision at any time. The **Prev** button is not available on the first record, and **Next** is not available on the last.

Records are ordered by date, then by station name. The tool always starts from the first record when you open a session.

5.6 The Image Panel

The right side of the screen displays the photograph(s) for the current record. If there are two images, they are shown side by side with a label above each one (IMAGE 1 and IMAGE 2) and the filename displayed at the top of the panel. Use the zoom buttons or double click on an image to zoom.

If an image cannot be found in the selected folder, a warning will appear in place of the photograph: “⚠ Not found” followed by the filename. In this case, check that the filename in the CSV matches the file in the folder exactly, and that the correct folder was selected. A record with no image entry, but with a tipping bucket reading will still be shown.

For split readings, all images from all sub-readings are displayed together in the image panel.

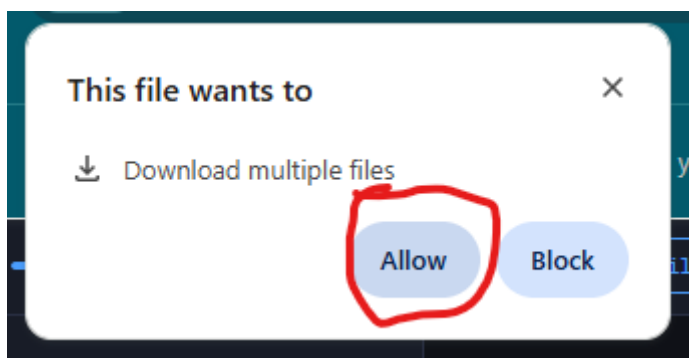
6. Ending a Session and Exporting Results

Click the **End & Export** button in the top-right corner at any time. You do not need to have reviewed all records before exporting.

A summary window will appear showing:

- The total number of records in the dataset
- How many you have reviewed so far
- The count for each decision type (MATCH, UNSURE, WRONG)
- How many records remain unreviewed

Click **Export All Files** to download the three output files. If you want to continue reviewing instead, click **Cancel** to close the window and return to the review screen. If you see the following message, click “Allow”:



Note: Unreviewed records are excluded from all output files. If you stop halfway through, you can continue from where you left off — but you will need to reload the files again from the setup screen, and your previous decisions will not be remembered. Do not close the tab without exporting – you will lose all your work!

6.1 Output Files

Three files are downloaded automatically to your default download folder when you click Export All Files:

File	Contents	Description
cleaned_data_YYYY-MM-DD_HH-MM.csv	Clean readings	Contains only the verified records labelled MATCH or WRONG (with your corrected values). Use this file for analysis. The timestamp in the filename shows when it was exported.
summary_table_YYYY-MM-DD_HH-MM.csv	QC summary	A table showing how many records were reviewed per station, and how many received each label.
training_notes_YYYY-MM-DD_HH-MM.json	Label archive	A detailed record of every decision you made, including corrected values, notes, and timestamps. This file can be used to train automatic image-checking tools in the future.

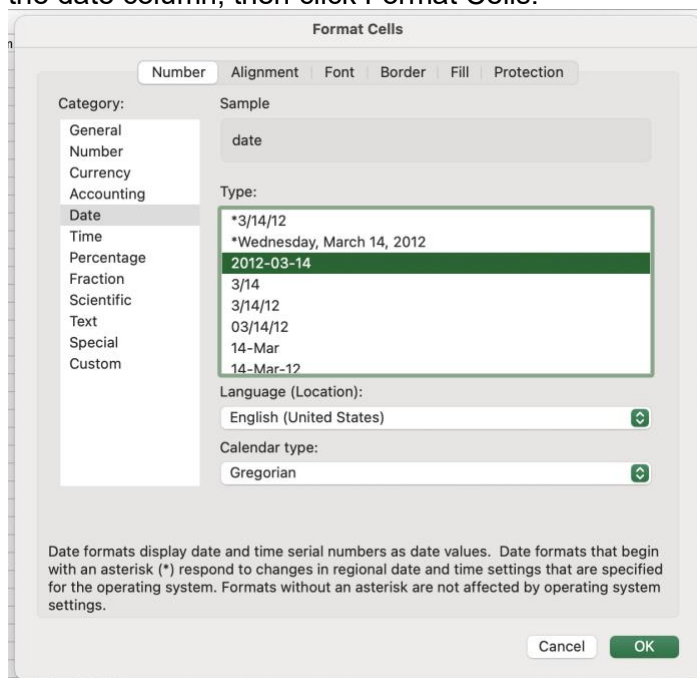
7. Troubleshooting

Q: The Start Review button does not activate

A: You must load at least the Gauge CSV and the Image Folder. Check that both show a green tick (✓).

Q: Nothing shows up in the review screen

A: Check that the date of your input csv files is in “YYYY-MM-DD” format. To change, highlight the date column, then click Format Cells.



Q: Images show “⚠ Not found”

A: The file name in the gauge CSV does not match the actual file in the folder. Check for: wrong capitalisation, missing .jpg extension, extra spaces, or the wrong folder selected. Re-selecting the correct folder requires reloading the page.

Q: The tool shows no records after loading

A: The gauge CSV may have no valid rows, or all rows may have been matched to bucket data only. Check that the gauge CSV has the correct column headers and at least one row of data with an image filename.

Q: The date column shows numbers instead of dates

Excel sometimes converts dates automatically to a number format. Use the formula `=TEXT (A2, "YYYY-MM-DD")` in a new column to convert it to the correct text format, then copy and paste the result as values before saving as CSV.

Q: The Confirm button does not respond

If you selected WRONG, make sure you have entered a value in the corrected value box before clicking Confirm.

Q: The tool opened in the wrong browser

Close the tab. Right-click the precip_checker.html file, choose Open with, and select Google Chrome.

Annex I: Preparing input file for totaling gauge readings

This section gives instruction on how to use an Excel macro which converts precipitation data into the format required.

The macro:

- Reads your data
- Extracts the required columns:
 - SubmissionDate
 - Location
 - rainfall_amount_dec
 - Image columns
- Creates a new file in the required format
- Saves the result as a new CSV file

No programming experience is required.

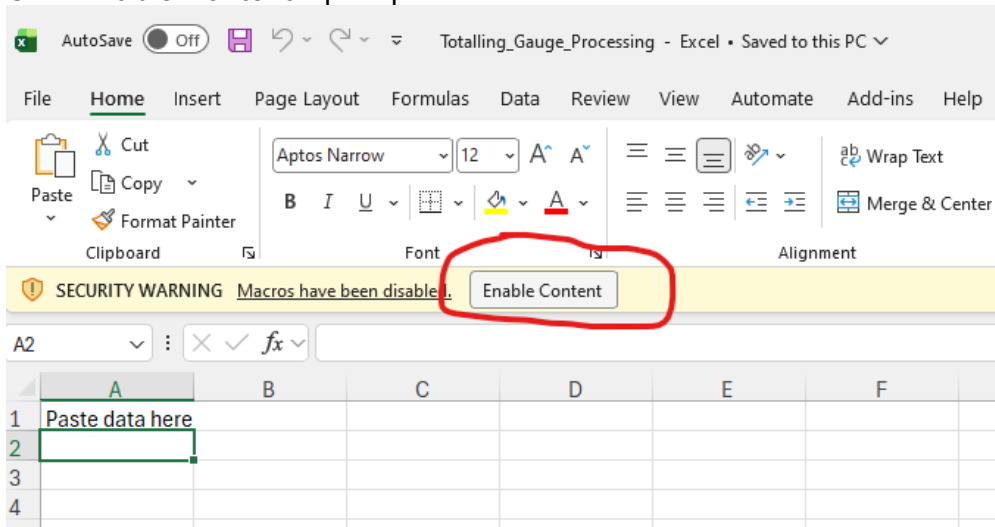
Step 1: Open the Totalling_Gauge_Processing.xlsm file

1. Double click the file to open in Excel.

Step 2: Adjust Macro Security Settings

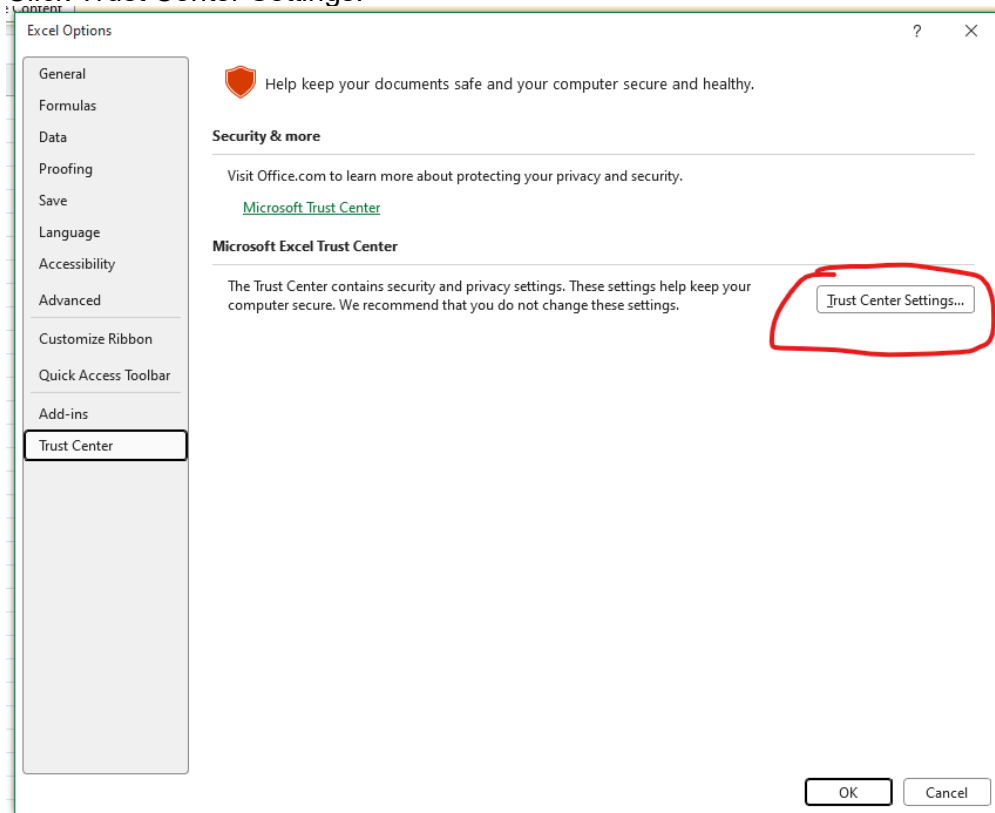
If macros are disabled, Excel may display a security warning.

1. Click **Enable Content** if prompted.



If you do not see the prompt:

1. Click File → Options.
2. Select Trust Center.
3. Click Trust Center Settings.



4. Select Macro Settings.
5. Choose Disable VBA macros with notification.
6. Click **OK** and reopen the workbook. You should see the message bar now. Click **Enable Content**.

Step 3: Open the ODK export file with gauge readings in Excel

1. Locate the CSV file with totaling gauge readings on your computer. The default file name from the ODK export is “TEMBO_Rainfall_Observatory.csv”.
2. Double-click the file to open it in Excel.
3. Verify that the data appears correctly in rows and columns.

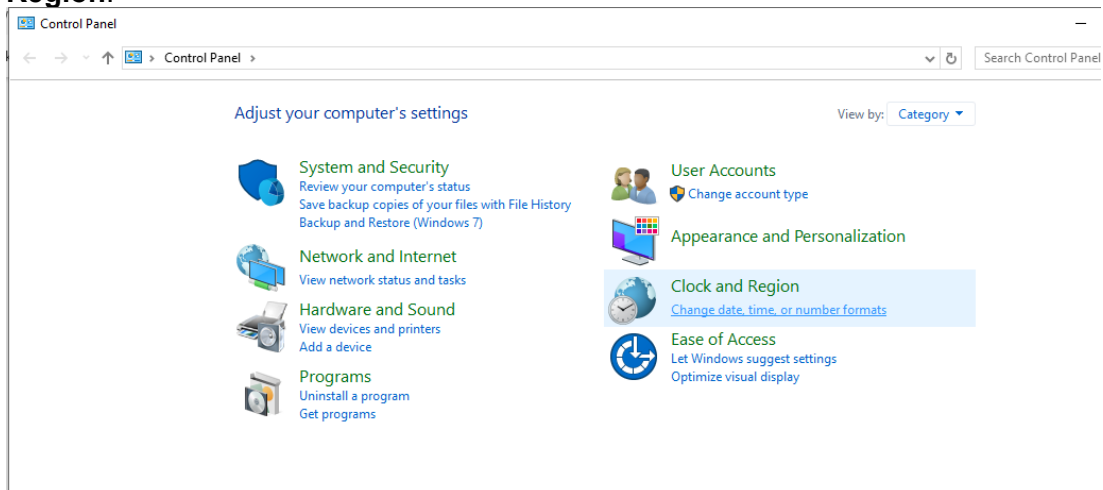
Correct, copy the data and go to [Step 4](#)

	A	B	C	D	E	F
1	SubmissionDate	user_full_name	location	time_recorded	rainfall_amount	image1
2	2026-06-11T08:32:15.036Z	Abass Amadu	Sanga	08:31:00.000	0	1781166
3	2026-06-11T08:01:40.344Z	Wunibiyeli mohammes	Kpalsogu	08:00:00.000	0	1781164
4	2026-06-11T06:45:19.775Z	Alhassan Seidu	Tuunaayili	06:30:00.000	0	1781160
5	2026-06-11T05:46:22.894Z	Yakubu Adam	Gbullung	05:45:00.000	0	1781156
6	2026-06-11T05:31:48.210Z	Ziblim A Fuseini	Garizegu	05:31:00.000	0	1781155
7	2026-06-10T07:02:16.601Z	Abass Amadu	Sanga	07:01:00.000	0	1781074
8	2026-06-10T06:31:50.652Z	Alhassan Seidu	Tuunaayili	06:30:00.000	0	1781072
9	2026-06-10T06:03:50.256Z	Yakubu Adam	Gbullung	06:03:00.000	0	1781071
10	2026-06-10T05:47:54.573Z	Alhassan Ibrahim	Kpaligu	05:47:00.000	20	1781070
11	2026-06-10T05:20:01.697Z	Ziblim A Fuseini	Garizegu	05:19:00.000	0	1781068
12	2026-06-09T17:55:59.980Z	Wunibiyeli mohammed	Kpalsogu	08:29:00.000	15	1780993
13	2026-06-09T12:43:00.000	Lovette Ranzurde warwies		12:43:00.000	11	1781009

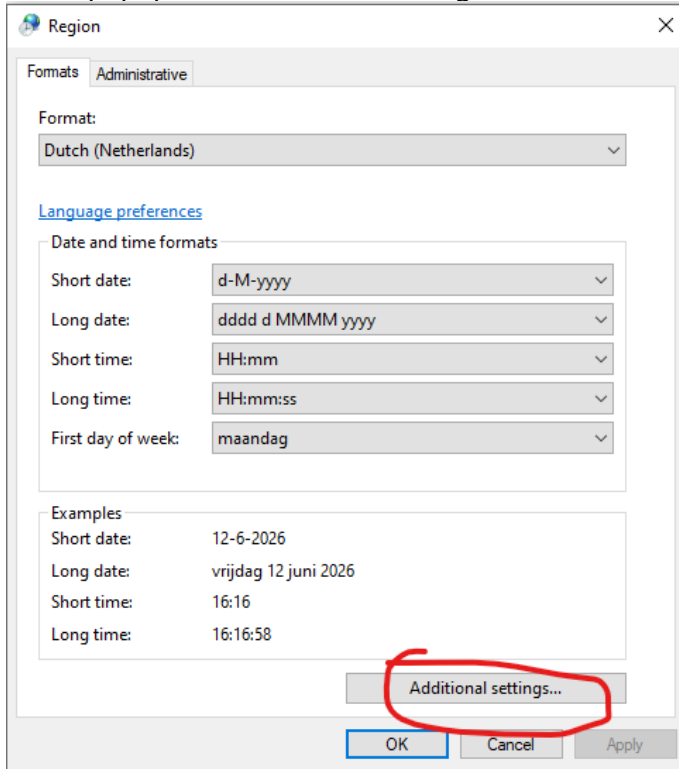
Incorrect, go to next bullet point

	A	B	C	D	E	
1	SubmissionDate	user_full_name	location	time_recorded	rain	
2	2026-06-11T08:32:15.036Z	Abass Amadu	Sanga	08:31:00.000Z		
3	2026-06-11T08:01:40.344Z	Wunibiyeli mohammes	Kpalsogu	08:00:00.000Z		
4	2026-06-11T06:45:19.775Z	Alhassan Seidu	Tuunaayili	06:30:00.000Z		
5	2026-06-11T05:46:22.894Z	Yakubu Adam	Gbullung	05:45:00.000Z		
6	2026-06-11T05:31:48.210Z	Ziblim A Fuseini	Garizegu	05:31:00.000Z		
7	2026-06-10T07:02:16.601Z	Abass Amadu	Sanga	07:01:00.000Z		
8	2026-06-10T06:31:50.652Z	Alhassan Seidu	Tuunaayili	06:30:00.000Z		
9	2026-06-10T06:03:50.256Z	Yakubu Adam	Gbullung	06:03:00.000Z		
10	2026-06-10T05:47:54.573Z	Alhassan Ibrahim	Kpaligu	05:47:00.000Z		
11	2026-06-10T05:20:01.697Z	Ziblim A Fuseini	Garizegu	05:19:00.000Z		
12	2026-06-09T17:55:59.980Z	Wunibiyeli mohammed	Kpalsogu	08:29:00.000Z		

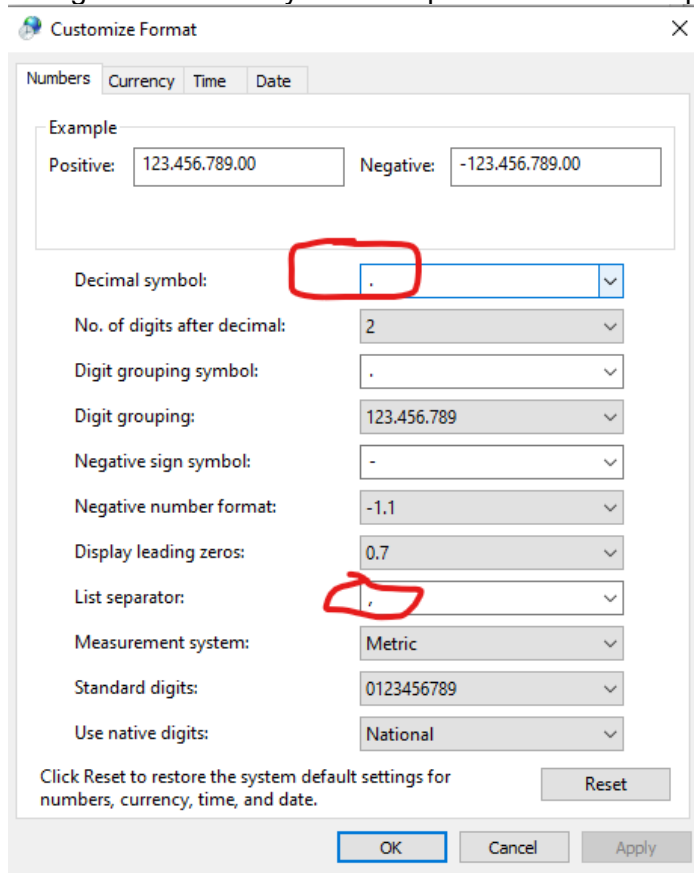
4. If they don't show up in columns, this is probably caused by the confusion that the comma “,” is a decimal, so Excel does not know how to separate data into columns. The easiest way is to change the regional settings on your computer. On your computer, go to **Control Panel**, then click **Change date, time or number formats** under **Clock and Region**.



5. In the popup, click additional settings.



6. Change the decimal symbol to a period “.” and list separator to a comma “,”.

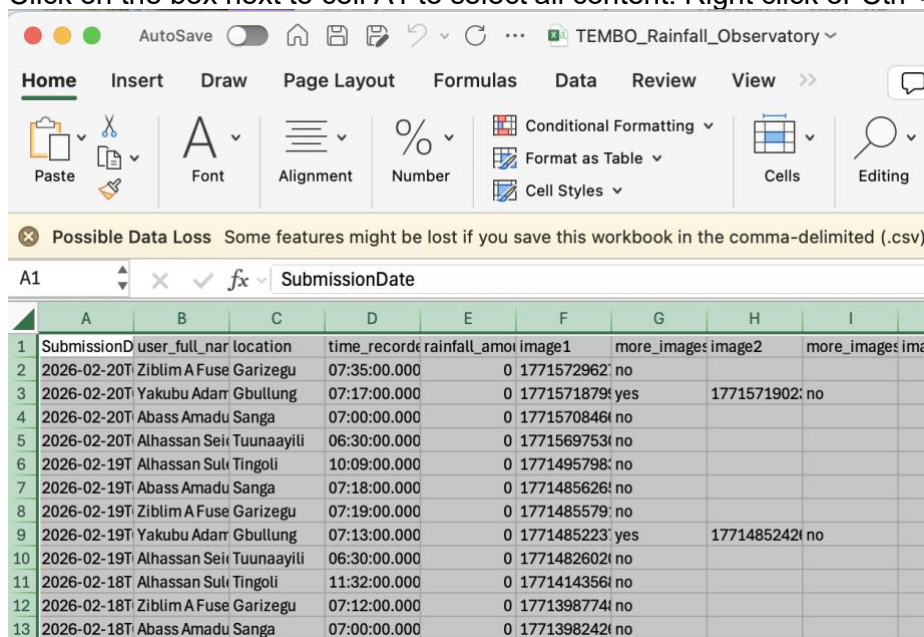


7. Close the .csv file and open it again. Check some rainfall values from the rain season. They should be displayed correctly.

832	2025-10-1	Alhassan S Tuunaayil	06:30:00.0	0	176025030	no		
833	2025-10-1	Yakubu A Gbullung	06:21:00.0	0	176025012	yes	176025013	no
834	2025-10-1	Augutine Golinga	07:02:00.0	6.6	1760166196876.jpg			
835	2025-10-1	Abass Am Sanga	07:00:00.0	16.3	176016608	no		
836	2025-10-1	Wunibiye Kpalsogu	08:48:00.0	15.5	1760172717897.jpg			
837	2025-10-1	Ziblim A F Garizegu	07:17:00.0	10.6	176016717	no		
838	2025-10-1	Alhassan S Tuunaayil	06:30:00.0	6.2	176016328	no		
839	2025-10-1	Yakubu A Gbullung	06:08:00.0	6.3	176016295	yes	176016300	no
840	2025-10-1	Wunibiye Kpalsogu	12:51:00.0	0	176010071	yes		
841	2025-10-1	Alhassan S Tingoli	11:34:00.0	0	176009607	no		
842	2025-10-1	Abass Am Sanga	07:15:00.0	0	176008058	no		
843	2025-10-1	Yakubu A Gbullung	07:15:00.0	0	176008054	yes	176008055	no

Step 4: Copy and paste all data to the Totalling_Gauge_Processing.xlsm file

1. Click on the box next to cell A1 to select all content. Right click or Ctrl + C to copy.

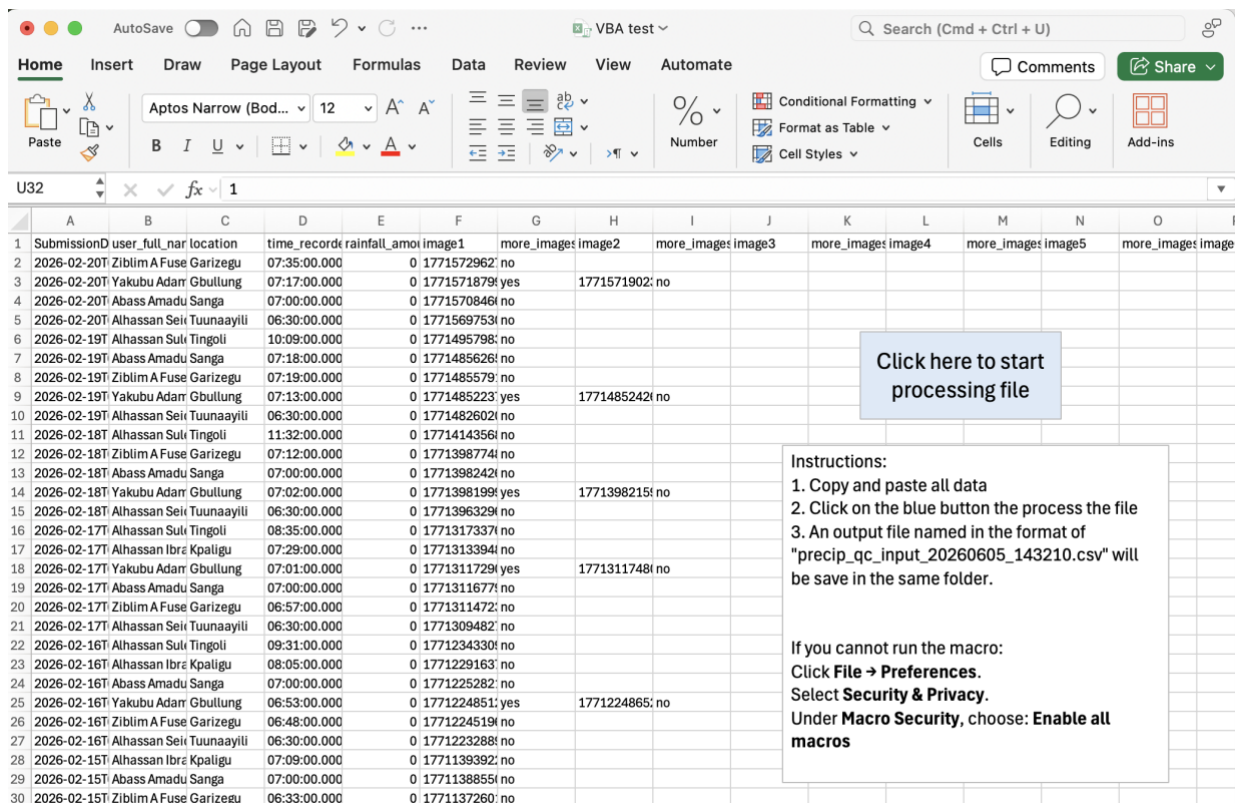


2. Go back to the Totalling_Gauge_Processing.xlsm file. Click on cell A1 and paste the data. You will see your copied data in the cell, with a blue button and white text box over it.

Step 5: Run the Macro

1. Click the blue button to run the macro
2. The macro will process the data.
3. If there is a popup window asking you to grant access for a specific folder, click on the **Select** button and select the folder requested.

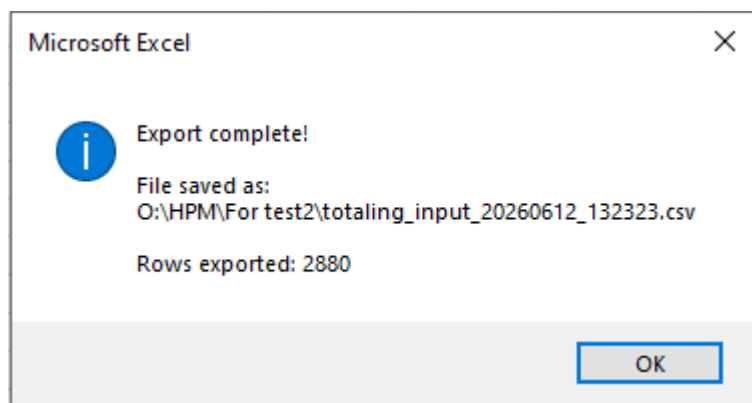
PrecipQC — User Manual



Step 6: Locate the File

When the macro finishes running:

1. A message will appear confirming that the export was successful.
2. The message will show the file location.



The file is usually saved in the same folder as the Excel workbook. This is the file you should select for [Section 4.1](#).

Annex II: Preparing input file for tipping bucket readings

This section explains how to convert raw tipping bucket logger data (exported from the logger software as an Excel file) into the `.csv` file required by the PrecipQC tool.

Two methods are provided. Method A uses a Python script that processes the data automatically. Method B achieves the same result using Microsoft Excel formulas only, with no programming required. Both produce identical output.

Method A: Python Script

This method runs a script that processes the Excel file automatically. It requires a one-time installation of Python and Visual Studio Code, which takes around 15–20 minutes the first time.

A1. Install Python

1. Open your web browser and go to: <https://www.python.org/downloads/>
2. Click the large yellow **Download Python** button. This downloads the installer for the latest version.
3. Run the downloaded installer. On the first screen, tick the box that says **Add Python to PATH** before clicking anything else. This box is near the bottom of the installer window. If you miss it, uninstall and reinstall.
4. Click **Install Now** and wait for the installation to finish.
5. Click **Close** when done.

Tip: To check Python installed correctly: press the Windows key, type cmd, press Enter. In the black window that opens, type `python --version` and press Enter. You should see something like Python 3.12.0. If you see an error, restart your computer and try again.

A2. Install Visual Studio Code

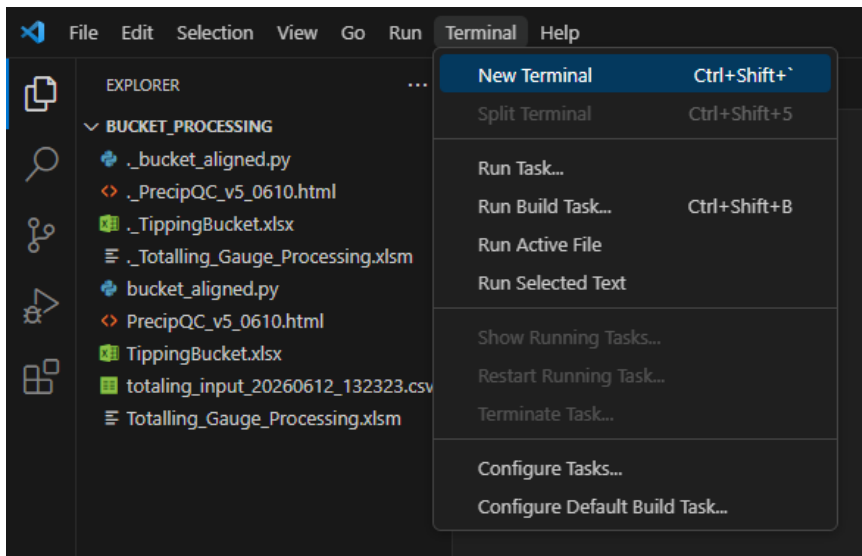
Visual Studio Code (VS Code) is a free text editor that makes it easy to run Python scripts.

6. Go to: <https://code.visualstudio.com/>
7. Click **Download for Windows** and run the installer.
8. Accept the default options throughout. On the “Select Additional Tasks” screen, tick **Add to PATH** if it is not already ticked.
9. Click **Install**, then **Finish**.
10. Open VS Code from the Start menu or desktop shortcut.
11. Install the Python extension inside VS Code:
 - a. Click the Extensions icon on the left sidebar (it looks like four squares).
 - b. In the search box, type `Python`.
 - c. Click the extension published by **Microsoft** and click **Install**.

A3. Install the Required Python Libraries

The script uses several Python libraries that are not included by default. You install them once using the Terminal inside VS Code.

12. In VS Code, open the Terminal by clicking **Terminal** in the top menu bar, then **New Terminal**. A panel opens at the bottom of the screen.



13. Type the following command and press Enter. Wait for it to finish (it may take a minute or two):

`pip install pandas openpyxl scipy numpy matplotlib xarray`

You should see something like this:

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS 0:\HPM\bucket_processing> pip install pandas openpyxl scipy numpy matplotlib xarray
Defaulting to user installation because normal site-packages is not writeable
Collecting pandas
  Downloading pandas-3.0.3-cp311-cp311-win_amd64.whl (9.9 MB)
    9.9/9.9 MB 57.5 MB/s eta 0:00:00
Collecting openpyxl
  Downloading openpyxl-3.1.5-py2.py3-none-any.whl (250 kB)
    250.9/250.9 kB 15.0 MB/s eta 0:00:00
Collecting scipy
  Downloading scipy-1.17.1-cp311-cp311-win_amd64.whl (36.6 MB)
    36.6/36.6 MB 59.5 MB/s eta 0:00:00
Collecting numpy

```

14. You should see a series of messages ending in *Successfully installed* (red box below). If you see an error saying `pip is not recognised`, try replacing `pip` with `pip3` and running again.

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  powershell + v [ ] [ ] ... ^ x

n-script-location.
WARNING: The scripts fonttools.exe, pyftmerge.exe, pyftsubset.exe and ttx.exe are installed in
'C:\Users\doriswong\AppData\Roaming\Python\Python311\Scripts' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-war
n-script-location.
Successfully installed contourpy-1.3.3 cycycler-0.12.1 et-xmlfile-2.0.0 fonttools-4.63.0 kiwisolver
-1.5.0 matplotlib-3.11.0 numpy-2.4.6 openpyxl-3.1.5 packaging-26.2 pandas-3.0.3 pillow-12.2.0 pyp
arsing-3.3.2 python-dateutil-2.9.0.post0 scipy-1.17.1 six-1.17.0 tzdata-2026.2 xarray-2026.4.0

[notice] A new release of pip is available: 23.1.2 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip
PS 0:\HPM\bucket_processing>

```

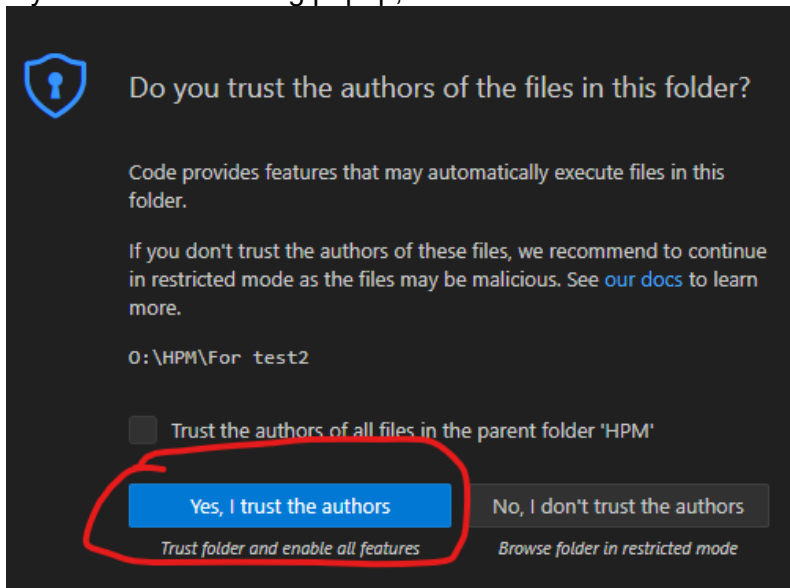
Note: You only need to do this installation once. The next time you use the script, go straight to Part A4.

A4. Prepare Your Files

15. Place your tipping bucket Excel file inside your working folder. Rename it to exactly `TippingBucket.xlsx` (the script looks for this exact filename).
16. Place your raw totaling gauge file inside this folder. This is often `TEMBO_Rainfall_Observatory.csv`. If it is not, change the name, since the script looks for this exact filename.
Important: Make sure this is the raw output file from the ODK containing submission times (not the processed file from Annex I).
17. Make sure the Python script `bucket_aligned.py` is inside the same folder.

A5. Open the Folder in VS Code and Run the Script

18. In VS Code, click **File** → **Open Folder** and select your working folder. Click **Select Folder**.
19. If you see the following popup, click on the blue button.



20. In the Explorer panel on the left, click `bucket_aligned.py` to open it.
21. Open the Terminal (**Terminal** → **New Terminal**).
22. In the Terminal, type the following and press Enter:

```
python bucket_aligned.py
```

23. Wait for the script to finish. When it is done, the terminal prompt will reappear with no error messages. If you see this in the terminal, you have successfully run the script!

	PROBLEMS	OUTPUT	DEBUG CONSOLE	TERMINAL
		location	date	bucket_mm
0		SANGA	2024-10-21	NaN
1		SANGA	2024-10-22	NaN
2		SANGA	2024-10-23	NaN
3		SANGA	2024-10-24	NaN
4		SANGA	2024-10-25	NaN
5		SANGA	2024-10-26	NaN
6		SANGA	2024-10-27	NaN
7		SANGA	2024-10-28	NaN
8		SANGA	2024-10-29	NaN
9		SANGA	2024-10-30	NaN
PS O:\HPM\bucket_processing>				

24. A new file called `bucket_aligned_{timestamp}.csv` will appear in your folder. Open `bucket_aligned_{timestamp}.csv` to check it looks correct. It should have three columns: `location`, `date`, `bucket_mm`. Each row is one station on one day. It should look like this on VS Code:

	<code>bucket_aligned_2026-06-12_15-01-09.csv</code>
1	location,date,bucket_mm
2	SANGA,2024-10-21,
3	SANGA,2024-10-22,
4	SANGA,2024-10-23,
5	SANGA,2024-10-24,
6	SANGA,2024-10-25,
7	SANGA,2024-10-26,
8	SANGA,2024-10-27,
9	SANGA,2024-10-28,
10	SANGA,2024-10-29,
11	SANGA,2024-10-30,

Scroll down to the rain season to check that there are days with rainfall, and that values make sense.

1523	GOLINGA,2025-09-09,
1524	GOLINGA,2025-09-10,
1525	GOLINGA,2025-09-11,
1526	GOLINGA,2025-09-12,22.4
1527	GOLINGA,2025-09-13,25.0
1528	GOLINGA,2025-09-14,2.2
1529	GOLINGA,2025-09-15,
1530	GOLINGA,2025-09-16,2.4
1531	GOLINGA,2025-09-17,5.4
1532	GOLINGA,2025-09-18,0.2
1533	GOLINGA,2025-09-19,34.4

25. If you want to check the intervals used, open the `intervals_export_{timestamp}.csv` file, which is also saved to your folder.

Tip: If you see an error message in the Terminal, read it carefully. The most common errors are: the Excel file is not named exactly `TippingBucket.xlsx`, or the file is open in Excel while the script is running. Close Excel and try again.

A6. Understanding the Output Columns

`bucket_aligned_{timestamp}.csv` has the following columns:

Term / Command	What it means
location	The station name, taken directly from the sheet name in your Excel file.
date	The calendar date of the reading, in YYYY-MM-DD format.
bucket_mm	Total rainfall in millimetres: number of tips × 0.2 mm per tip.

You are now ready for [section 3](#) of the manual.

Method B: Microsoft Excel Only

This method does not require Python or any programming. It uses Excel formulae and built-in tools to produce the same output. It takes longer to set up, is more prone to manual errors but requires no software installation. **The file will have a universal cutoff at 7am**, regardless of the reading time of the totaling gauge, making it less accurate.

Important: This method works best if your logger data has the same structure across all station sheets. If the column names or layouts are very inconsistent between sheets, the Python script (Method A) will handle this more reliably.

B1. Understand Your Source File

Your Excel file (`TippingBucket.xlsx`) has one sheet per station. On each sheet, there is a column for the timestamp (when the tip occurred) and a column for the event value (1 = real tip, 0 = other). The column names may vary slightly between sheets (e.g. “Time/Date” or “Time”; “Event” or “Event Type”).

Before starting, open the file and note down:

- The exact name of the timestamp column on each sheet (e.g. “Time/Date”)
- The exact name of the event column (e.g. “Event”)
- Which row the data starts on (usually row 2, after a header row)

B2. Prepare Each Station Sheet

Work through every station sheet one at a time. For each sheet, you will add three columns: the station name, an adjusted date, and a valid tip flag.

Add the station name column

1. Find an empty column to the right of your data. Label it location in the header row.
2. In the first data row, type the station name exactly as you want it to appear in the output (e.g. Tuunaayili).
3. Copy that cell down to all data rows. The quickest way: click the cell, copy it (Ctrl+C), then select all the rows below it in that column and paste (Ctrl+V).

Add the adjusted date column

4. In the next empty column, add the header `adjusted_date`.
5. In the first data row, enter the following formula, replacing B2 with the cell containing the timestamp for that row:

```
=INT(B2 + 17/24)
```

This adds 17 hours to the timestamp, which shifts the 7 AM morning cutoff to midnight. Tips recorded before 7 AM are thereby grouped with the previous calendar day, matching the observer's morning reading routine. `INT()` then strips the time portion, leaving only the date.

6. Format the `adjusted_date` column as a date: select the column, right-click → **Format Cells** → **Custom** → type YYYY-MM-DD in the box → click OK. The cells should now show dates like 2025-06-02. If they still show numbers (e.g. 45695), the format has not been applied correctly — repeat this step.
7. Copy the formula down to all data rows.

Add the valid tip column

8. In the next empty column, add the header `valid_tip`.
9. In the first data row, enter the following formula, replacing C2 with the cell containing the event value for that row:

```
=IF(C2=1, 1, 0)
```

10. Copy the formula down to all data rows.

Repeat for all stations before moving to the next step. Every sheet should now have three new columns: `location`, `adjusted_date`, and `valid_tip`.

B3. Combine All Sheets Into One

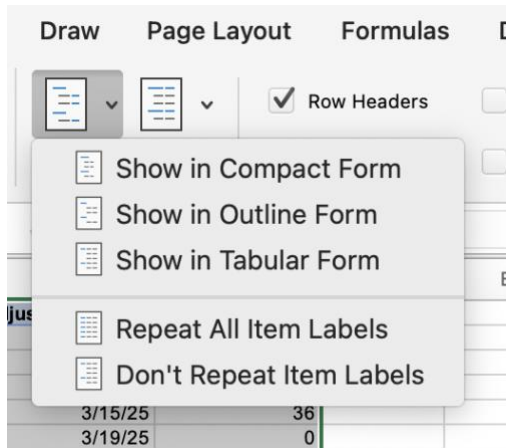
You will combine all the data into a single sheet first, then build one Pivot Table covering all stations at once.

1. Add a new sheet to the workbook and name it `all_data`.
2. Go to the first station sheet. Select all the data including the header row (click the top-left cell, then press Ctrl+Shift+End to extend the selection to the last used cell). Copy it (Ctrl+C).
3. Go to the `all_data` sheet and paste into cell A1.
4. Go to the second station sheet. Select all the data **excluding the header row** (start from row 2). Copy and paste it into the `all_data` sheet, immediately below the last row of data already there.
5. Repeat for all remaining stations. The `all_data` sheet should end up with one header row at the top and all tips from all stations below it, with the `location` column identifying which station each row belongs to.

B4. Build One Pivot Table for All Stations

1. Click anywhere inside the data on the `all_data` sheet.
2. Go to **Insert** → **PivotTable or PivotChart** → select relevant columns → choose **New Worksheet** → click OK.
3. In the PivotTable Fields panel on the right:
 - Drag `location` to the **Rows** area.

- Drag `adjusted_date` to the **Rows** area, below location.
 - Drag `valid_tip` to the **Values** area. Make sure it says **Sum of valid_tip**. If it says **Count**, click the field → **Value Field Settings** → choose **Sum**.
4. Remove station-level subtotals:
 - Right-click any station name in the pivot table, → select **Field Settings**.
 - In the dialog, go to the **Subtotals & Filters** tab, → under Subtotals, select **None**, → click **OK**.
 5. Enable tabular layout and repeat station names:
 - With the pivot table selected, go to **the PivotTable Analyze** tab (or Analyze or Design tab) in the ribbon.
 - Click the **Report Layout** dropdown button, → choose **Show in Tabular Form**.



- Click Report Layout again, → choose Repeat All Item Labels.
6. Verify dates are ungrouped:
 - If you see months/years instead of individual dates, right-click any date in the pivot table, → select **Ungroup**.
 7. Click Refresh (in the PivotTable Analyze tab) to ensure all data and settings are up to date.

	A	B	C
1	location	Adjusted_date	Sum of Valid_tip
2	Galinkpegu	2/24/25	1
3	Galinkpegu	3/6/25	2
4	Galinkpegu	3/13/25	8
5	Galinkpegu	3/15/25	36
6	Galinkpegu	3/19/25	0
7	Galinkpegu	4/10/25	3
8	Galinkpegu	4/11/25	47
9	Galinkpegu	4/17/25	18
10	Galinkpegu	4/19/25	1
11	Galinkpegu	4/23/25	2
12	Galinkpegu	4/27/25	1
13	Galinkpegu	4/28/25	8
14	Galinkpegu	4/29/25	213
15	Tingoli	2/7/25	23
16	Tingoli	3/19/25	2
17	Tingoli	4/11/25	63
18	Tingoli	4/18/25	11

Figure 4. End result after step B4

B5. Build the Output Table

1. Copy the entire Pivot Table result — the location names, dates, and tip sums. Do not include the grand total row if one appears.
2. Paste as values only into a new sheet named `combined`: right-click the target cell → **Paste Special** → **Values** → OK. This removes the Pivot Table formatting and leaves plain data.
3. Rename the columns to match the required output format. Your four columns should be:

```
location | date | tips | bucket_mm
```

4. The `bucket_mm` column does not come from the Pivot Table — you need to add it. In the first data row of the `bucket_mm` column, enter:

```
=C2*0.2
```

Replace C2 with the cell containing the tip count for that row. Each tip equals 0.2 mm of rainfall, so this formula converts the tip count to rainfall in mm. Double click the green square at the bottom right of the cell to autofill the formula to all rows below.

5. Check that the date column is formatted as YYYY-MM-DD. If it shows numbers (e.g. 45695), select the column → right-click → **Format Cells** → **Custom** → type YYYY-MM-DD → OK.
6. Sort the data by date, then by location: go to **Data** → **Sort**. Add two levels — first by date (A to Z), then by location (A to Z).

B6. Save as CSV

1. Click the `combined` sheet tab to make sure it is the active sheet.
2. Ctrl + A to copy all cells of the file. Open a new Excel workbook, and paste the cells you copied.
3. In this new workbook, go to **File** → **Save As**.
4. Name the file `bucket_input.csv`.
5. In the Save as type dropdown, select CSV (Comma delimited) (*.csv).
6. Click **Save**. Excel might warn you that CSV cannot save multiple sheets — click **OK** or **Keep Current Format**.
7. Open the saved file in Notepad (right-click → Open with → Notepad) to verify the first few lines look like:

```
location,date,tips,bucket_mm
Garizegu,2025-06-02,12,2.4
Tisunayili,2025-06-02,5,1.0
Sanga,2025-06-03,3,0.6
```

Important: If the date column appears as a number in the CSV (e.g. 45695), go back to the Excel file, re-format the date column as custom YYYY-MM-DD, then re-save as CSV. Excel sometimes loses date formatting when saving to CSV.

Tip: To quickly check whether all dates are in the correct format, open the CSV in Excel and look for any cells that appear right-aligned (numbers) in the date column. Dates stored as text will be left-aligned.

You are now ready for [section 3](#) of the manual.